

Game Theory and Social Choice

Lecture 7: Games in Time

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Games That Take Time

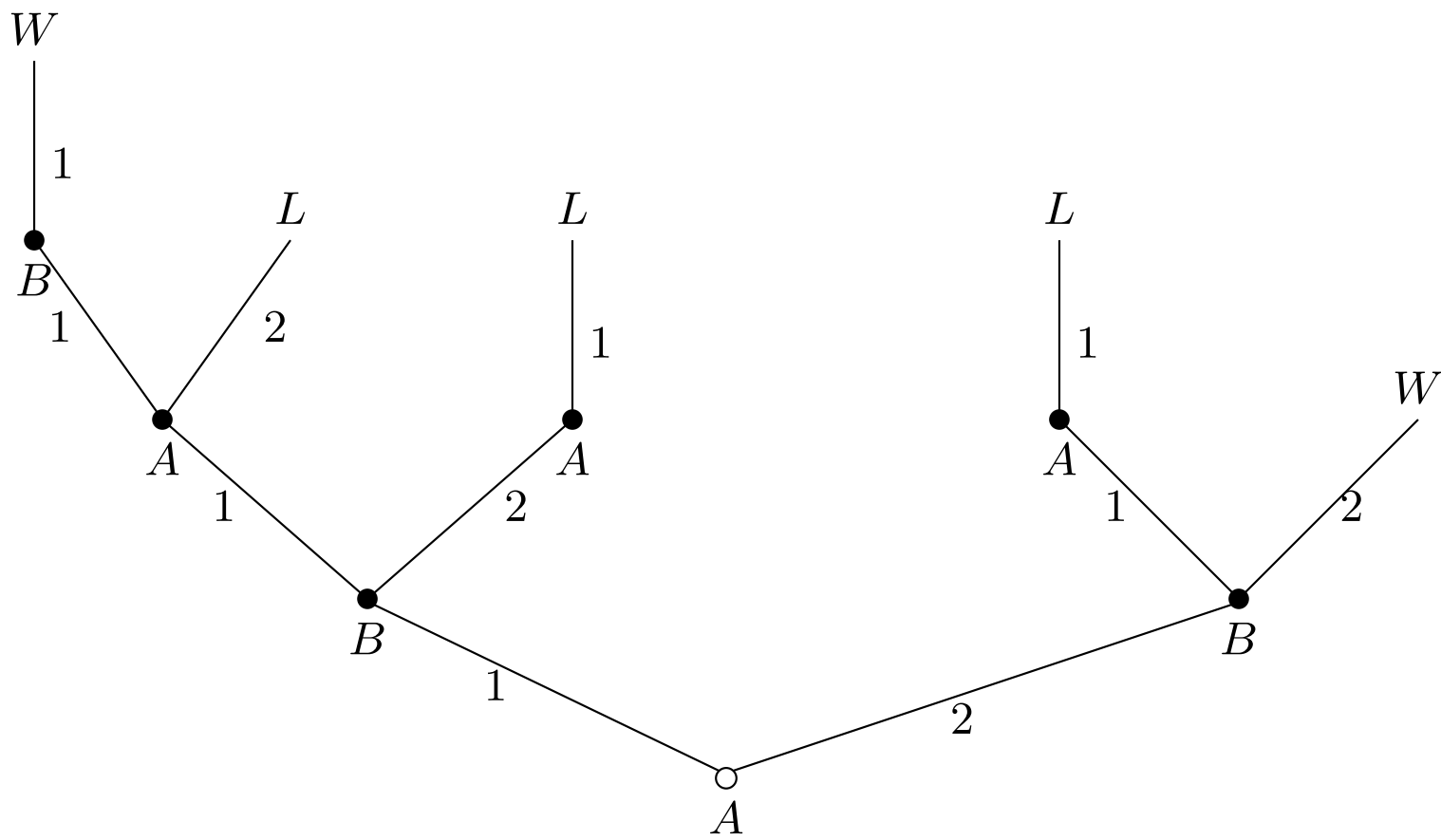
A Very Small Game

Last Counter

Four counters sit on the table. A and B take turns removing counters, A moving first. On your turn you must take one counter or two. Whoever takes the **last** counter loses.

Before we do anything formal: would you rather move first or second?

The Extensive Form



Reading the Tree

Start at the node that isn't filled in. Here that's at the bottom.

A moves first, taking 1 or 2. Then B , and so on until the counters run out. The number on each branch is how many were taken.

At the top is who won. W means A wins, L means B wins. Since this is zero-sum, one label per outcome is enough.

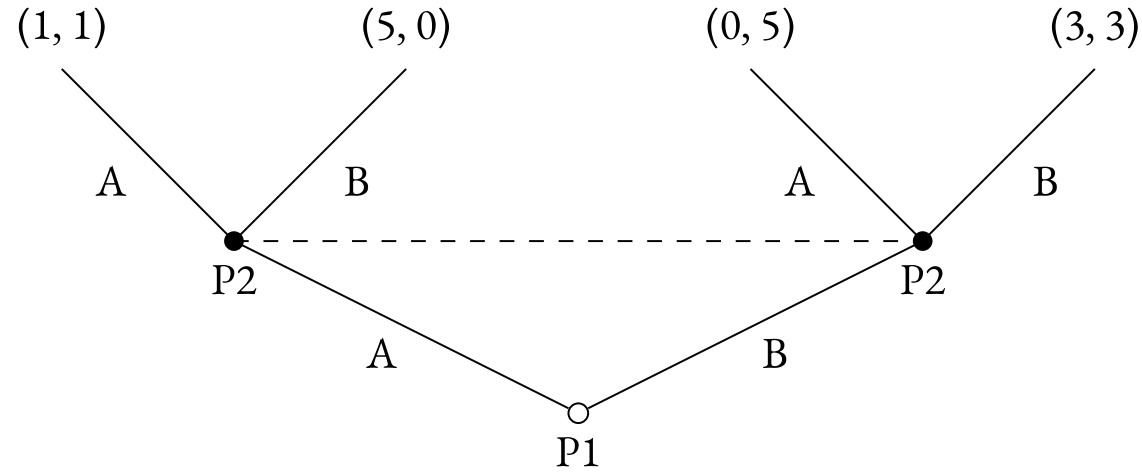
Reading the Tree

Two things worth noticing. The branches are not all the same length: the game can be over after two moves or run to four. And some nodes have a single branch out of them, because with one counter left there is nothing to decide.

We can generalise this beyond two players, or to games where who moves at a point of time is a function of which moves were made earlier; it's a fairly flexible form.

Simultaneous Moves, In a Tree

We can even represent simultaneous move games this way.



Information Sets

The dashed line there means P2 does not know which of those two nodes they are at when they move. The connected nodes form an **information set**.

We only connect nodes where the same player moves and the same moves are available.

An unmarked node is in an information set of its own.

So we can draw simultaneous games as trees. Player 1 “moves first”, but Player 2 learns nothing from it.

An Assumption In the Syntax

Drawing lines between nodes, or a circle around the nodes like you see in some books, is only a good model of P2's uncertainty if make a really strong assumption.

It assumes that at any node in the set, the open nodes are the same.

Epistemic Possibility

Say that node y is possible at node x if when a player is at x they could be, for all they know, at y . In the form we're using, there's a line between x and y , or a circle around them.

The syntax forces the following two assumptions.

Transitivity

If y is possible at x , and z is possible at y , then z is possible at x .

Symmetry

If y is possible at x , then x is possible at y .

Philosophers debate **Transitivity**, and almost uniformly reject **Symmetry**, but they are both accepted here.

Strategies

Strategies

A **strategy** is a plan for what to do in any circumstance whatsoever.

Not just “what I will do”, but “what I would do at every node where I might have to move”.

A moves at four nodes: the opening, the node with two counters left, and two different nodes with one counter left. At a node with one counter left there is nothing to choose.

So *A*’s strategy has two free parts: her opening, and what to do if two counters are left when it’s time to move again. Write it $\alpha\beta$.

Counting

B moves at three nodes, and again only two of them offer a choice: three counters left, or two. Write B 's strategy as $\delta\epsilon$.

So each player has $2^2 = 4$ strategies. We'll basically ignore 'forced' moves here.

Note that A 's strategy specifies what to do after moves the strategy itself rules out.

This turns out to matter, and it will not quite cause complications, but make it possible to state some complications.

The Normal Form

Table 1: Last Counter in normal form, A 's rows against B 's columns

	11	12	21	22
11	W	W	L	L
12	L	L	L	L
21	L	W	L	W
22	L	W	L	W

What the Table Shows

Look at B 's column 21: take 2 if three counters are left, take 1 if two are. Whatever A does, A loses.

Now look for a row of all W 's. There isn't one. A has no winning strategy at all. Of course, if there is a column of all L 's, there couldn't be a row of all W 's.

So the *second* player wins this game. This is easier to see on the table than the tree.

What The Table Hides

The tree has five outcomes. The table has sixteen cells.

Each outcome turns up several times. Change one outcome on the tree and you have to change many cells in the table.

Worse, A 's last two rows are identical. That's because strategies 21 and 22 differ only at a node that A 's own opening move makes unreachable.

Backwards Induction

Working From the End

Start where there is nothing to decide.

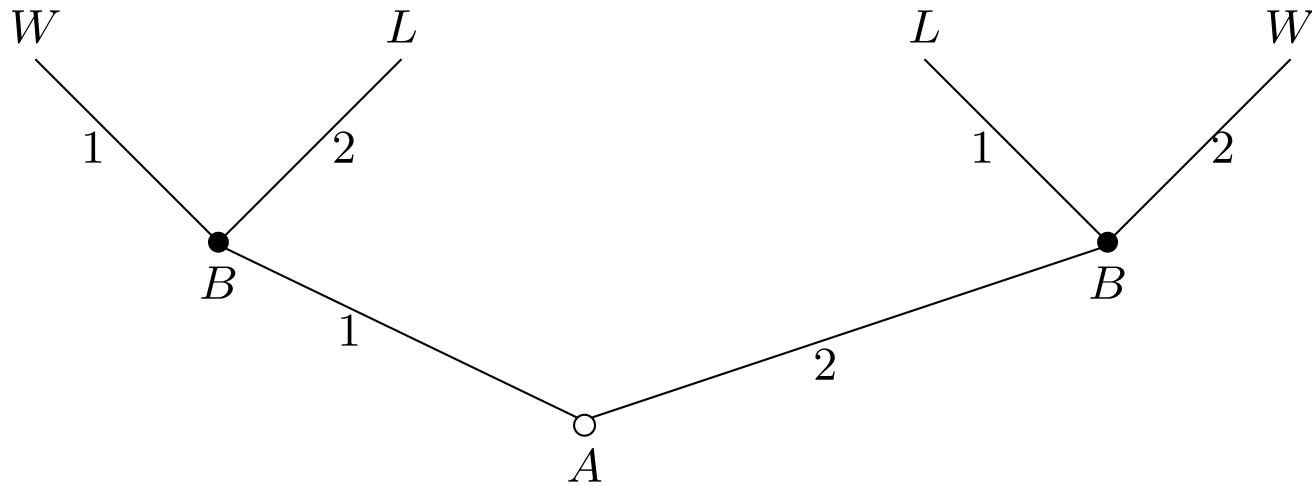
If you face one counter, you must take it, and taking it loses. So whoever leaves exactly one counter has already won.

That settles three branches of the tree immediately.

One Step Back

Now A 's one real late choice: two counters left, her move. Taking 1 leaves a single counter for B , so she wins. Taking 2 loses on the spot. She takes 1.

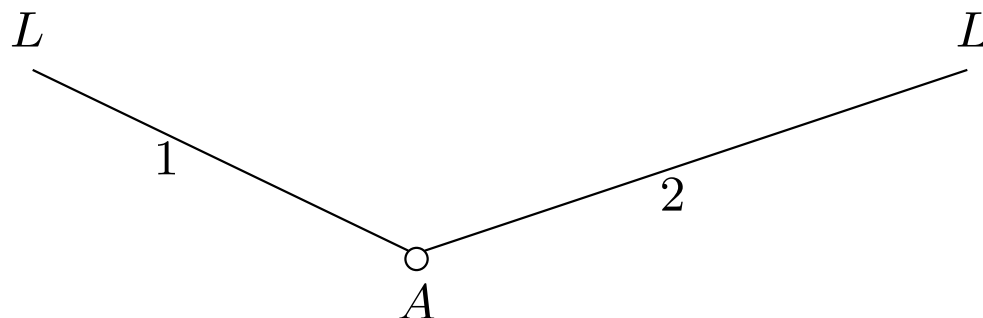
Redraw the game with everything after B 's move already settled.



Two Steps Back

Now do the same to B . B wins when A loses, so B heads for L .

If A opened with 1, three counters remain and B takes 2. If A opened with 2, two remain and B takes 1. Either way, B leaves one counter.



The Answer

Both of A 's openings lead to L . A cannot win.

B 's strategy is 21

The Answer

leave exactly one counter

The Answer

which is exactly the column of L s we found in the table.

Try It

i Exercise

Play the same game with xix counters, same rules. Who wins, and what is the winning strategy?

Hard Question

i Discussion

Then suppose A opens badly from six. B is now standing at a node that a perfectly rational A would never have created. Should B go on assuming that A is rational?

Hidden Assumptions

Backwards induction requires a lot of background.

Each player is rational. Each knows that. Each knows that the other knows it. And so on, for at least as many rounds as there are moves in the game.

Even more strongly, those assumptions have to be **modally robust**.

When we did the analysis, we started at nodes without asking how we got there, or whether we could have rationally got there. We acted like everyone would be rational there, and expect rationality, no matter what happened before that. This seems questionable.

A Different Game

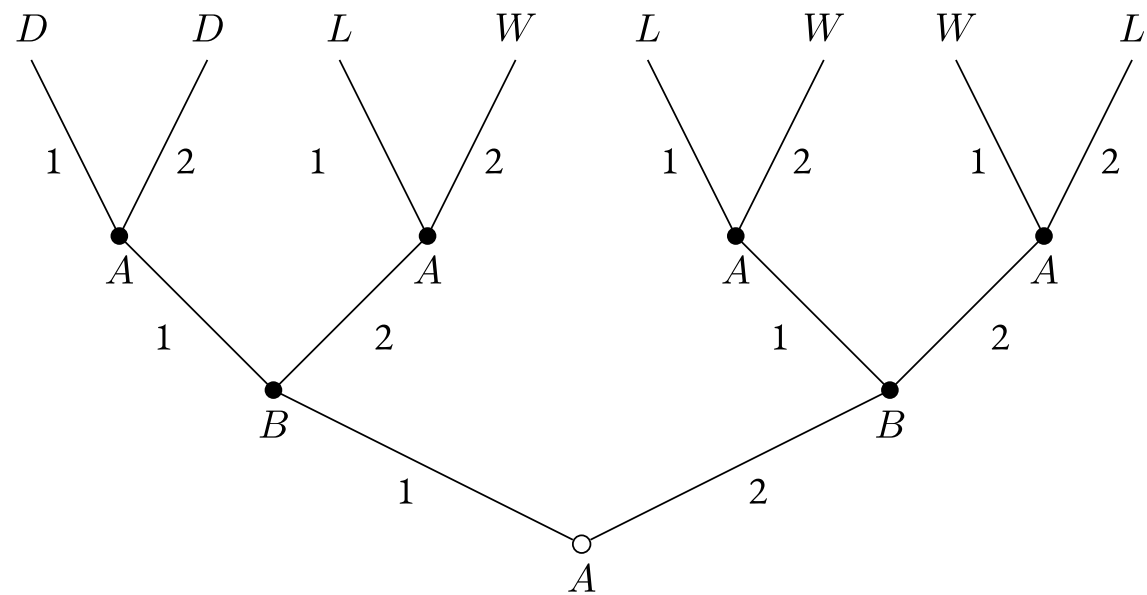
💡 Five

A moves, then B , then A again. Each move announces 1 or 2. A wins if the three numbers sum to 5; B wins otherwise.

This one is worked through in full in the notes at 5.2, tree and normal form alike. I'm flagging it here to make a point about a variant of it.

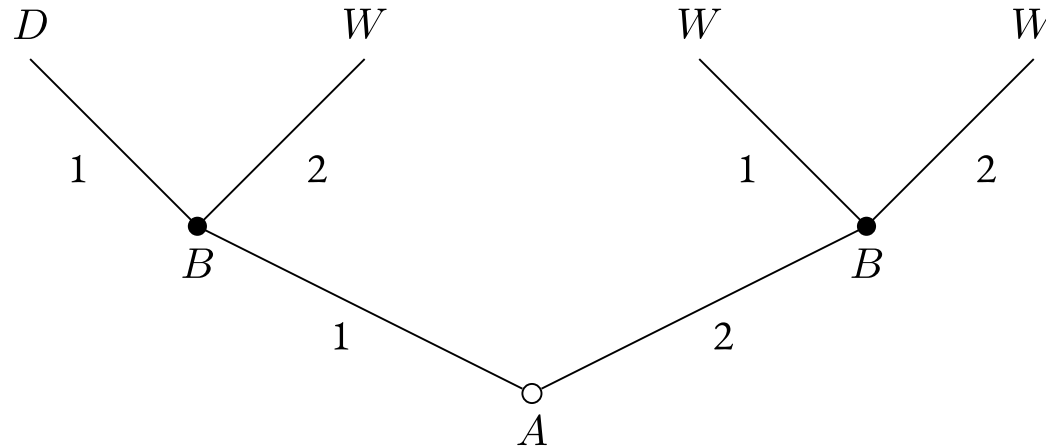
The Variant

Five, except that the two leftmost outcomes are draws rather than wins for B .



The Problem

Assume A plays optimally at the last move, and the top row collapses.



Suppose A opens with 1. Now B chooses. Playing 1 gets a draw; playing 2 lets A win. So B takes the draw.

But Why Believe That

The claim that playing 2 leads to A winning came from assuming A is perfectly rational.

But if A were perfectly rational, she would have opened with 2 and given herself a forced win.

B is standing at a node that only gets reached if A is **not** playing perfectly. Why should she keep assuming perfect rationality from here on?

Taking the draw is probably still wise. But the argument that playing 2 *is* choosing defeat looks broken.

Why This Matters

In a normal form game, this never comes up. You never get evidence that the other player is an idiot before making your one and only move.

In a tree, how much confidence you have in the other player keeps coming up. This is a big deal.

The Value of a Game

The Setup

Let's focus for a bit on games with the following features:

- ▶ Alternating moves
- ▶ The full history of the game is public, so no hidden moves
- ▶ There is a known cap on how many moves are possible
- ▶ Zero-sum

There are finitely many outcomes, so order them by A 's preferences. Say A 's least preferred outcome has value 0, the next 1, and so on.

Defining Value

Define the **value** of a node recursively.

- ▶ A **terminal** node has the value of its payoff.
- ▶ A node where A chooses has the **greatest** value among the nodes she can move to.
- ▶ A node where B chooses has the **least** value among the nodes she can move to.

The **value of the game** is the value of the initial node.

This is an important application of backwards induction.

Subgames

A **subgame** is what you get by treating some node as an initial node and keeping everything downstream.

The node has to be one where it is common knowledge that we have arrived.

So nodes inside a non-degenerate information set do not start subgames; you cannot play a game without knowing which game it is.

A subgame is a game, in the way a subset is a set.

Three Results

1. The value of a game is the value of one of its terminal nodes.
2. A can guarantee an outcome **at least** the value of the game.
3. B can guarantee an outcome **at most** the value of the game.

All three go by induction on the length of the game. They are in the notes at 5.3.

What Follows

Suppose rationality is common knowledge, and stays that way throughout.

Then the outcome will be exactly the value of the game. A makes sure it is not lower, B makes sure it is not higher.

So these games are, in a precise sense, already decided before anyone moves.

Chess and Checkers

Both satisfy every condition on the list. The only one that isn't obvious is finiteness, but the draw rules ensure it. (Question for class: why is this so?)

Each ends in White win, Black win, or draw. So one of those three is the value.

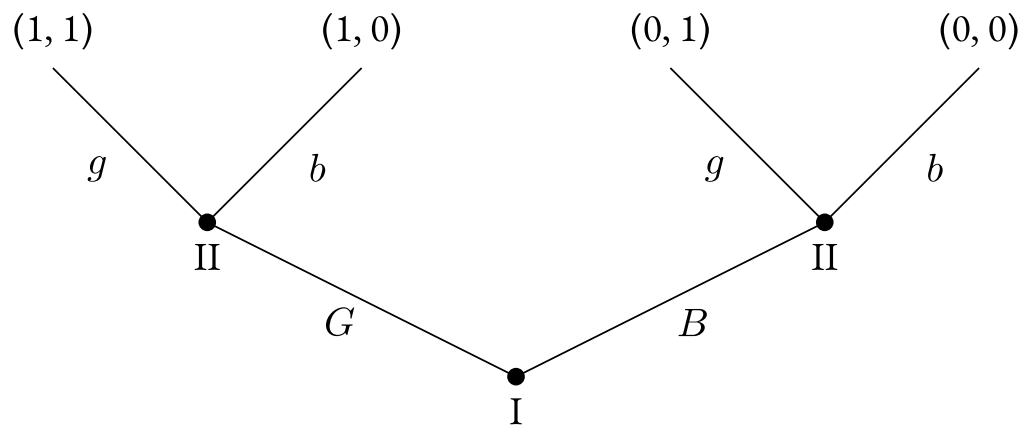
For checkers we know: it's a draw, proved in 2007 by the Chinook project at the University of Alberta. The reference is in the notes at 5.3.

Chess is probably also a draw, but nobody has proved it, and given the size of the tree nobody is about to.

Subgame Perfect Equilibrium

Easy Game

Player I moves, then Player II. Each gets 1 for choosing Good and 0 for choosing Bad.



Its Normal Form

Player II's strategy is $\alpha\beta$: what she does if I plays Good, then what she does if I plays Bad.

Table 2: Easy Game in normal form

	gg	gb	bg	bb
G	1, 1	1, 1	1, 0	1, 0
B	0, 1	0, 0	0, 1	0, 0

There are two Nash equilibria: G with gg , and the bolded one, G with gb .

The Odd One

G with gb is a Nash equilibrium. Neither player can do better given what the other does.

But it is strange. Player II is planning to take 0 at a node where 1 is freely available.

It is true that gb is weakly dominated by gg . Keep that in mind; this is a version of the point we made earlier that Nash permits weakly dominated strategies.

Removing the Bad Equilibrium

Look at the subgame starting at Player II's right-hand node.

It is not much of a game. Player I has no choices; Player II picks 1 or 0.

But it is a game, and in it, b is **strictly** dominated. So gb is not a Nash equilibrium of that subgame.

The Definition

Subgame Perfect Equilibrium

A strategy pair is a **subgame perfect equilibrium** when it is a Nash equilibrium, and it is a Nash equilibrium of every subgame.

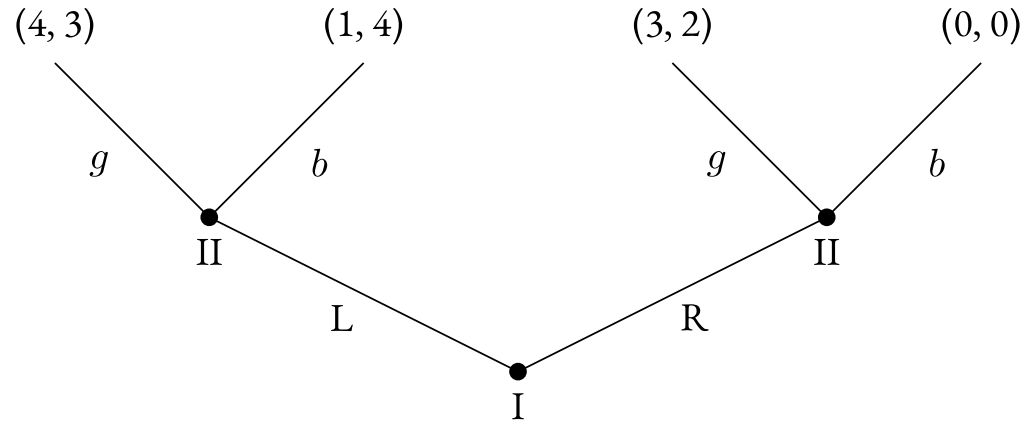
G with gb does not satisfy this test, because b is not a Nash equilibrium of a subgame.

G with gg passes the test, and is the only one equilibrium of the game.

In perfect information games, backwards induction finds subgame perfect equilibria.

Different Outcome Equilibria

In Easy Game the two equilibria pay the same, so nothing much turned on it. That need not happen.



Incredible

Table 3: Incredible in normal form

	<i>gg</i>	<i>gb</i>	<i>bg</i>	<i>bb</i>
L	4,3	4,3	1,4	1,4
R	3,2	0,0	3,2	0,0

There are two Nash equilibria, both bolded. Backwards induction gives *R* with *bg*.

But let's talk about the other one, *L* with *bb*.

The Incredible Threat

If Player II is committed to bb , then Player I's best reply really is L .

So Player II would like Player I to believe she will play b on the right.

But look at what that means when we play the game out. At the right-hand node she would be choosing 0 when 2 is right there.

Now in this equilibrium we never reach that node, so she never has to follow through. She's bluffing, and the bluff never gets called.

What We Are Requiring

Subgame perfection says that is not good enough.

It requires players not merely to make choices that make sense, but to be **disposed** to make choices that make sense, everywhere, including at nodes that will never be reached.

That is a strong demand. But there is something plausible to the idea that we shouldn't accept bluffs like this, or what the literature calls **incredible threats**, or sometimes **non-credible threats**.

Next Time

We have been assuming players keep reasoning correctly all the way down the tree.

Next time we'll ask what happens in the more realistic case where we quibble about that.